Blockchain Security for the Internet of Things

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1 Introduction

Internet of Things (IoT) has numerous IoT devices such as cameras, sensors, mobiles, smart refrigerators, and smartwatches. IoT devices face many challenges when transferring data from one IoT device to another. IoT devices are attacked or misused by the hacker or attacker devices made to secure data transfer using Blockchain Technology. Blockchain innovation is the disappeared connect to determine protection and unwavering quality uncertainties on the IoT. Blockchain is the silver shot required by the IoT business. It very well may be utilized in following billions of associated gadgets, permitting the handling of exchanges and harmonization between gadgets. This takes into consideration critical investment funds for IoT industry makers. This decentralized approach would crash weak links, making a heartier biological system for the gadgets. The cryptographic method utilized by blockchains would make customer information more classified. The principal benefits of the Blockchain are extraordinary straightforwardness, upgraded security, further developed detectability, high proficiency, low costing, and no outsider intercession. The IoT empowered Blockchain, and very much planned motivations will

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motivate to replace consumer information [1]. Many facets of today's markets are likely to be impacted by the large-scale adoption of the blockchain method.

Blockchain Technology provides security and privacy, for example, Bitcoin [2]. The Internet of Things (IoT) speaks to one of the essential troublesome improvements of this century [3]. It is a characteristic development of the Internet (of PCs) to the inserted and cyber-physical frameworks, "things" that, while not PCs themselves, by and by have PCs inside them. IoT exponential growth in research and industry, however contempt, all experiences protection and security vulnerabilities [4]. Systematic security and protection methods will generally be inapplicable for IoT, for the most part, because of its decentralized topology and the asset limitations of most of its gadgets. The Blockchain supports digital money [5].

IoT Blockchain Platform Architecture IoT blockchain platform has four layers: (1) application layers, (2) IoT blockchain service layer, (3) connectivity layer, and (4) physical layers. Figure 1 shows the IoT Blockchain Platform Architecture.

Application Layers The application layer contains an application program that can communicate with IoT devices and perform a particular task.

IoT Blockchain Service Layer This helps the layer contain all modules that put together standard administrations to give different provisions of blockchain innovations like Identity management, consensus, and peer-to-peer management. The dispersed record is an agreement of recreated, shared, and synchronized computerized information spread across the blockchain network, where network individuals claim the duplicate of the record. It gives extra safe room to record the gadget organization and detect information given by actual sensors. Any progressions to the record are reflected in all duplicates in the blockchain network in practically no time. The record can be either permissioned or premise.

Connectivity Layer The primary function provided by the connectivity layer is routing management. Self-association is required since actual gadgets themselves have no worldwide web conventions (IPs). This layer likewise incorporates different modules for giving underneath administrations: network the administrators, secure the administrators, message representative, and route the administrators.

Physical Layer This contains an immense quantity of IoT devices, user devices, data storage, local bridges, and servers that are linked together around a peer-to-peer blockchain network.

Blockchain technology became well-known with the emergence of cryptocurrency mining. However, it now has a lot to do with the Internet of Things. The quantity of data handled by IoT devices is massive. It is all sent in a chain, leaving it vulnerable to hackers. In this context, the opportunity to use the Blockchain architecture to verify, standardize, and safeguard the adoption of data handled by devices presents itself. For IoT security, the Blockchain can track the data gathered by sensors and prevent it from being replicated by erroneous data. To secure the

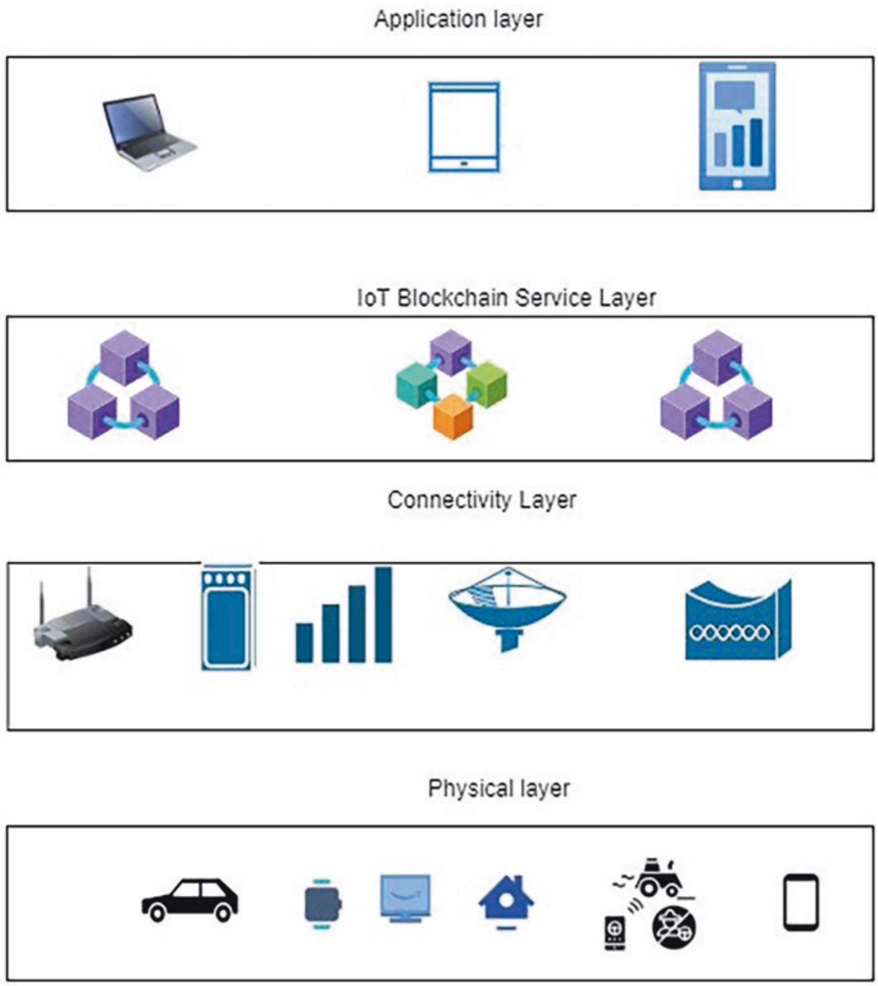

Fig. 1 IoT Blockchain platform architecture

end-to-end integrity of the gathered data and related interactions, IoT applications require trust mechanisms that cut across these layers. Transparency of data collecting procedures and related interactions, as well as the ability to audit these processes and interactions, are critical to meeting these criteria. The needs for openness and auditability encourage the use of blockchain to support IoT trust. Blockchain is immutable because network participants control it together using a consensus method such as Proof-of-Work (PoW) and Proof-of-Stake (PoS). Inter-node interactions and data collection are both part of the Internet of Things. Existing research treats the IoT trust challenge as two distinct issues, proposing reputation and trust techniques. For IoT data collection or inter-node communication, there is one. As a result, there is a clear requirement for an integrated architecture that provides end-to-end trust across data gathering and blockchain. In the Internet of Things, node interactions are important. In Blockchain-based IoT applications, layered trust architecture is used.

This book chapter is organized as follows: Sect. 2 applies blockchain security and privacy for the Internet of Things. In Sect. 3, we provide the conclusion of the blockchain applications in a secure manner.

2 Related Works

This section aims to discuss the blockchain applications in different areas and address the significant security challenges and overcome the breaches. Blockchain has a different application. In this paper, we have mentioned the following list of applications: (a) Supply Chain and Logistics, (b) Insurance, and (c) Society Application [6].

2.1 *Supply Chain and Logistics*

Goods exchange between supplier and consumer is a complex task because Blockchain is comfortable for Supply Chain and Logistics [7] and is transparent to the customer. It gives assurance to the companies who send the product to the customer. Before, they were given a clear audit trail to manage the risk and reduce the supply chain failures; now, in this digital advertising, using Blockchain implemented the IoT devices as follows:

Figure 2 shows the supply chain management using a blockchain-enabled IoT device. Blockchain Technology provides accurate and end-to-end tracking of supply chain management systems [7]. Supply chain management can be done using two different tag types: (1) NFC (Near Field Communication) and (2) RFID (Radio Frequency Identification) (Table 1).

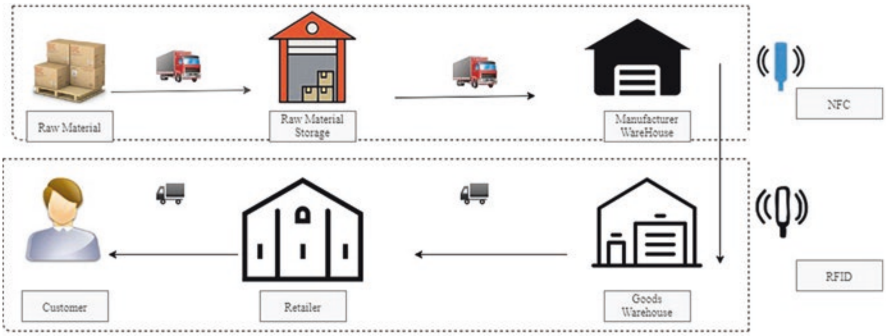

Fig. 2 Supply chain management using blockchain-enabled IoT device

Table 1 Comparison between RFID and NFC

Parameter	RFID	NFC
Network Type	Point – Point	Point – Point
Communication	Unidirectional	Bidirectional
Range	Up to 100 m	More than 0.2 m

The raw material is transferred to raw material storage using NFC, and it is used for short communication. When any goods are exchanged, it will create a blockchain node that contains information about the excellent exchange. That node is also added to the distributed ledgers. Raw material storage to manufacturer is also exchanged using NFC. Goods warehouse to retailer and retailer to customers are exchanged using an RFID tag. An RFID tag is used for long-distance communications. Blockchain technology characteristics such as decentralization and immutability improve supply chain management [8].

2.2 Insurance

Traditional Insurance Policies are the expected protection arrangements that are regularly handled on paper contracts, which means cases and installments are mistakes inclined and regularly require human supervision. Modern systems are considered as a kind of distributed ledger in the Blockchain. The distributed ledger is one of the significant advantages of blockchain technology. It improves the efficiency of fraud elimination and data analysis with the Internet of Things [9]. Exchanges are attached in a dispersed framework on the system, which makes framework recuperation by disposing of a solitary purpose of disappointment or incorporated element, during the distributed ledger in the insurance application to enable the IoT smartphone devices to access the Insurance policy in the distributed ledger [1, 10]. The best five protection giants mutually forced blockchain Insurance Industry Activity (B3I) to examine the application practicality of Blockchain in the protection industry [11] and create blockchain-based verifications of ideas for protection in the proofs of concept for Insurance [2, 12–14] (Fig. 3).

2.3 Society Applications

Society Application is a significant concern in blockchain technology. In feature generation, we have to solve the traditional borrowing relationship that reduces the credit problems. So we have to change the smart contract. Trust and security are the significant advantages of the intelligent contract [15].

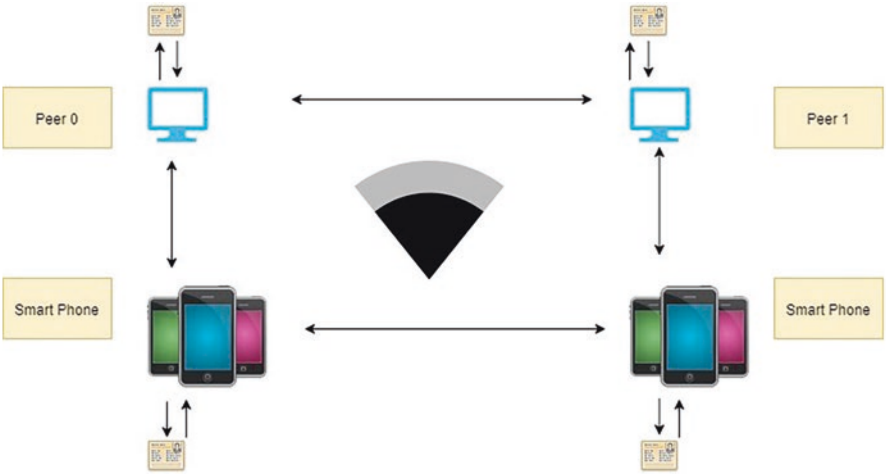

Fig. 3 Distributed ledger using IoT smartphone device

2.4 Blockchain Music

In traditional music recording in the film industry, the copyright amount is not claimed for the Musician's music because it has directly gone to the copyright seller. This problem never occurs when using blockchain secure manner; this should be implemented in this music era [9, 11]. For example, copyright sellers give the royalty to the Musician and the buyer of the music got and used it in concerts. Figure 4 depicts the music seller's copyright using a blockchain smart contract to the buyer and gives the royalty amount to the appropriate Musician [6]. The benefits of this smart contract are the easiest way together with the Blockchain and speed and accuracy, trust, savings, and security. The digital artwork and healthcare industry as well as diamond industry are the other applications used in the smart contract [11].

2.5 Energy Financing

Blockchain will simplify raising capital for the perfect vitality ventures using interfacing progressively potential financial specialists [16]. Now, India also launched a smart city [17]. To develop each state and mainly concentrate the green energy field, the cryptocurrencies are utilized [18] (Fig. 5).

This book chapter provides the blockchain application in various organizations, and the individual domain uses blockchain technology. Blockchain is also used in education, transportation and charity. We have described the comparison between blockchain technology and IoT applications.

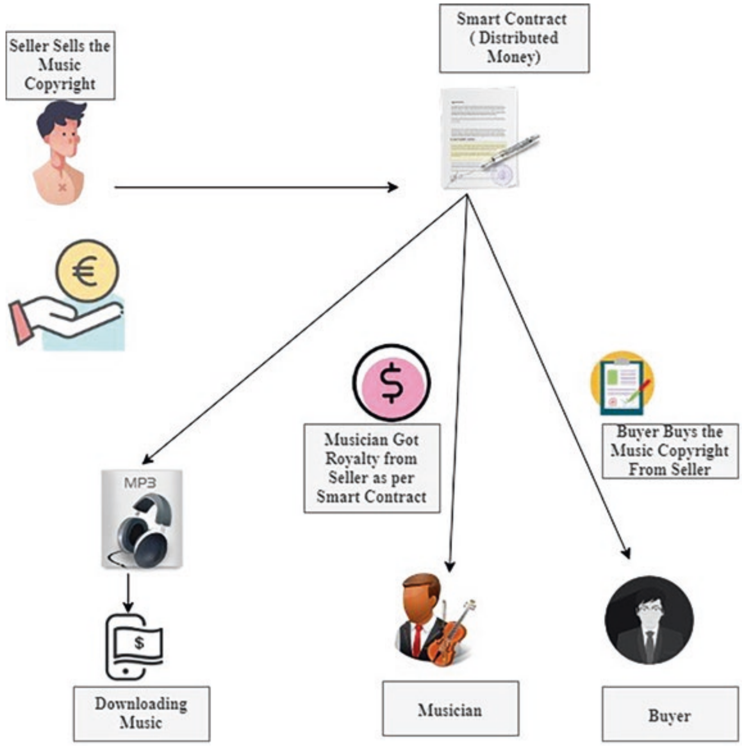

Fig. 4 Music copyright using smart contract

Fig. 5 Green energy using blockchain technology

2.6 Automotive Industry in IoT and Blockchain

Digitization is turning into a serious necessity. The auto business is utilizing IoT-empowered sensors to construct completely independent vehicles. Associating Industrial IoT arrangements in the car area to a decentralized organization permits various clients to share basic data quickly. The car area is an appealing blockchain IoT use case, as the joined innovation can disturb, for example, gasoline installment frameworks, autonomous vehicles, and automated traffic light with clever leaving. NetObjex has shown how Blockchain and IoT might be utilized to make a smart stopping framework. For acknowledging ongoing vehicle and distinguishing the accessibility of parking spots, the firm has joined forces with a leaving sensor startup called "PNI." The association smoothes out the most common way of discovering a parking space and computerizes installments utilizing digital currency wallets. Stopping time costs are determined utilizing IoT sensors, and paying is done straightforwardly through the crypto-wallet [19].

In the case of carsharing, blockchain technology may be able to provide a secure digital identification. Personal preferences and settings may be safely kept in the automobile via blockchain, preventing exposure to third parties. This might potentially make peer-to-peer carsharing possible. Meanwhile, consumers may benefit from blockchain since a single registration in the blockchain ecosystem may be utilized for all carsharing offers from various service providers in the ecosystem.

Purchasing or selling a vehicle: For automobile owners, blockchain-based registries would make it easier to verify a vehicle's history (for example, if it has been in an accident), allowing for greater openness when buying a car. Another advantage would be having a comprehensive picture of the vehicle's components: users could research the origins of the carpeting and solve repair-related issues, not to mention that a smart contract allows the seller and buyer to enforce a products' transaction without the use of an intermediary. Vehicle finance and leasing: We were able to improve and automate several procedures in the car leasing and finance sector, thanks to blockchain-based smart contracts. Deactivating the unlocking system, for example, can prohibit an automobile from being utilized if the lease fee has not yet been paid.

2.7 Smart Home Using Blockchain and IoT

Smart IoT-enabled gadgets are becoming increasingly important in our daily lives. The Internet of Things (IoT) blockchain enables home security systems to be controlled remotely from a smartphone. The typical centralized way of exchanging data created by IoT devices, on the other hand, lacks security requirements and data ownership. By addressing security concerns and eliminating centralized infrastructure, Blockchain has the potential to take smart homes to the next level. Each bright house in British Columbia is built for a unique purpose. Devices produce store transactions in order to store data. An SP or the house owner generates an access transaction. Make use of cloud storage: A homeowner or a service provider creates a transaction for monitoring to watch a gadget regularly. It is possible to add a new gadget to the smart home. A device is removed through a genesis transaction and a deletion of the transaction. All of the deals mentioned above have been completed. To keep communication safe, a shared key is utilized. Lightweight to identify any changes in transactions, hashing is used. Biometrics, voice recognition, and face recognition are examples of sensitive user data kept on the Blockchain for better

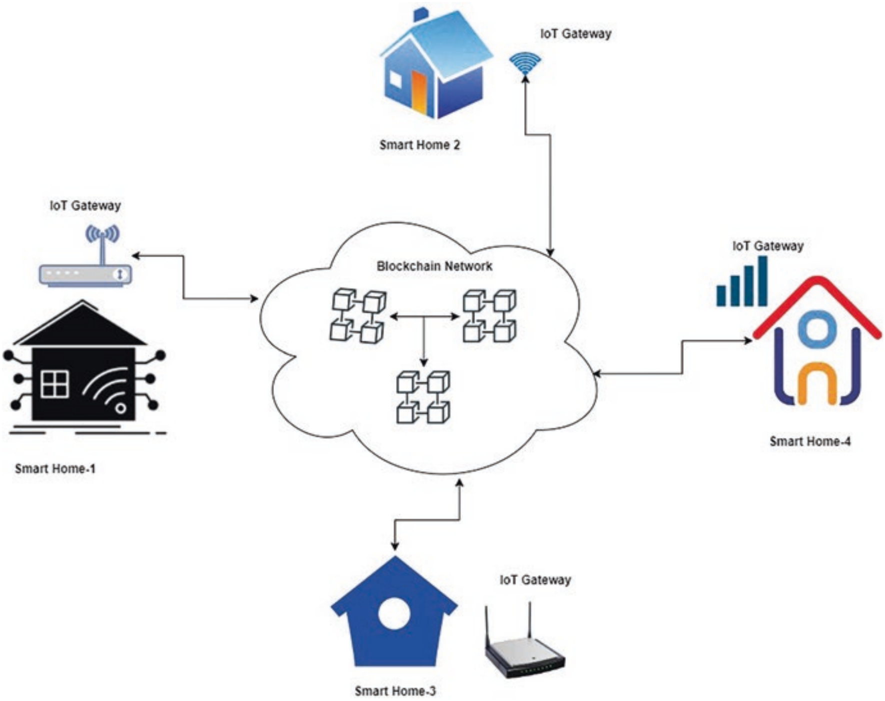

Fig. 6 Smart home management using IoT and Blockchain

security. Once data is recorded on the Blockchain, it cannot be changed, and only the proper people have access to it. Figure 6 depicts the overview of innovative home management using Blockchain and IoT.

2.8 Pharmacy Industry

The problem of counterfeit medications in the pharmaceutical industry is getting worse day by day. The pharmacy industry is in charge of medication development, manufacture, and distribution. As a result, tracing a drug's whole route is challenging. The transparent and traceable characteristics of blockchain technology can aid in tracking medicine shipments from their point of origin to their final destination in the supply chain.

Medi ledger is a blockchain IoT use case for tracking the lawful transfer of ownership of prescription medications. When it comes to monitoring delicate healthcare goods, transparency and traceability are critical. Manufacturers, distributors, dispensers, and end-customers may access data recorded on the distributed ledger, which is immutable and timestamped. Medi ledger is a blockchain-based technology that allows consumers to regulate their access and prevent counterfeit medications from entering the supply chain.

2.9 Agriculture Industry

Agriculture industries are increasing the use of farmland to generate the best feasible yields in order to profit and meet human food demands. Growing more food for the growing population while reducing environmental impacts and maintaining supply chain transparency is critical for optimum consumer satisfaction. The combination of Blockchain and IoT can completely transform the food production business, from farm to grocery store to house. Installing IoT sensors in farms and transmitting the data straight to the Blockchain can improve the food supply chain. Figure 7 depicts the overview of Blockchain and IoT use cases in the agriculture industry.

Data received from farms utilizing smart agricultural instruments is extremely valuable since it aids in the making of profitable decisions. Because there is no

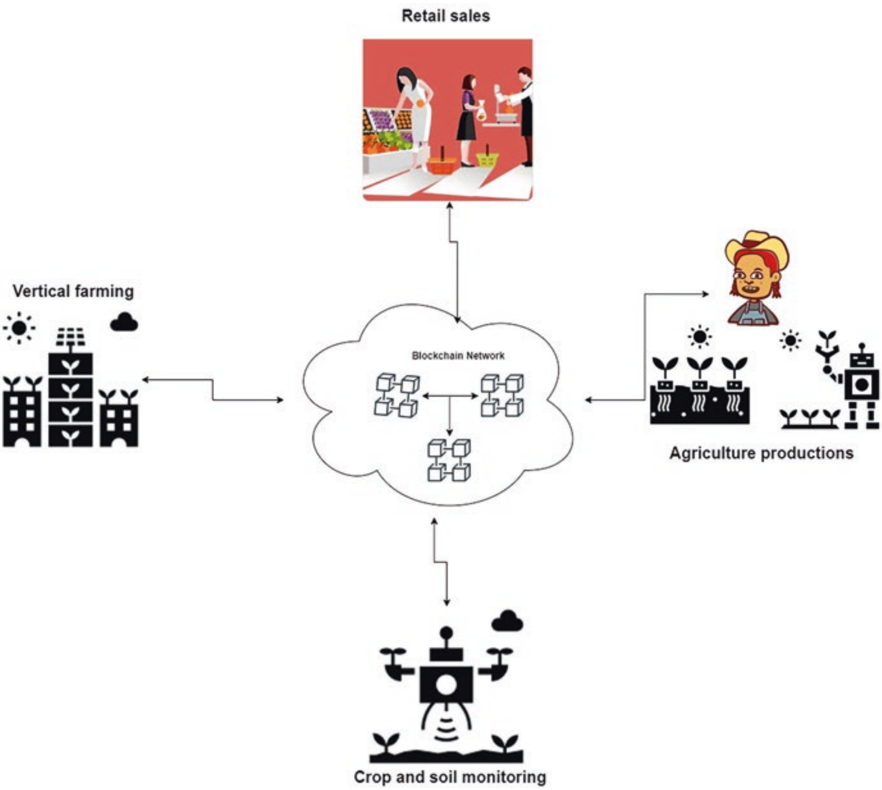

Fig. 7 An overview of Blockchain and IoT use case in the agriculture industry

industry standard for handling agricultural data, producers have a tough time doing so. Many farmers and producers are unaware of how to effectively use data for decision-making. As a result, it's critical to provide farmers and producers with the tools and strategies they need to successfully gather, organize, process, and apply data.

2.10 Water Management

Distribution of water resource to the end users is complicated because water management has traditionally relied on outdated technology and convoluted methods that may no longer be applicable today. Water the board project has been proposed to utilize Blockchain and IoT in order to evaluate stream contamination. Libelium and Airalab have worked together on a task called "Robot on the Volga." It gathers water contamination levels independently by utilizing a robot equipped with IoT sensors and blockchain innovations. The robot takes water readings from the Volga River's Kuybyshev Reservoir. He communicates the information on the Ethereum Blockchain continuously. The robot can utilize IoT to sort out where and when estimations were gotten, helping researchers in deciding the wellspring of contamination. It can screen and save information by means of NetObjex's IoT and blockchain administrations (Table 2).

2.11 IoT and Blockchain Security Challenges

Combining Blockchain with IoT leads to a lot of potential problems, which will enable hackers to steal the information present in the IoT edge devices. Blockchain-enabled identity access management provides security to the IoT edge devices. IoT devices developers and IoT services providers face many security challenges when they want to develop secure IoT devices for the end users using blockchain technology. The security challenges are discussed in these sections [20–23]. Figure 8 depicts the security challenges of Blockchain and IoT.

2.11.1 Overly Large Attack Surface

IoT devices has large attack surface if you consider smart home IoT application hackers can steal the information from any devices attached to smart home. The managing of IoT devices is through the Internet, and more prominent quantity of administrations might be assaulted. The attack surface is the term for this. One of the main stages during the time spent ensuring a framework is to lessen the assault surface. There might be open ports on a gadget with administrations running that aren't really essential for working. An attack on a particularly futile assistance

Table 2 IoT based Blockchain applications and challenges

S.No	Blockchain applications	Challenges	Advantages
1.	Supply chain and logistics	Lack of transparency Advertisements fraud	Using NFC and RFID Tag gives transparency to the Seller and Consumer. Can monitor the goods easily so no possible fraud.
2.	Insurance	Traditional insurance is the paper contract, and brokers' challenges occur.	Distributed ledgers are used to improve efficiency. IoT smartphones are also used to eliminate the fraud/brokers in the insurance sector.
3.	Society application	Traditional borrowing has some credit problems.	Implement the intelligent contract to improve the accuracy and trust between the lender and borrower.
4.	Blockchain music	Traditionally musician cannot get royalty to the particular seller in the music industry.	Implement the blockchain technology so the seller can sign the intelligent contract using music industry, the musician can get royalty, and the buyer can get the copyright from a seller without any risks.
5.	Energy financing	Internal (investors) and external workflows have not been fulfilled in the energy field.	It is easier to raise the capital/smart city with clean and green energy by utilizing the funds.
6	Automotive industry	The rise of autonomous vehicles and the evolution of connected vehicles	A blockchain-based system that allows automakers to identify automobiles with problematic parts and, as a result, issue-customized recalls or service advisories for such vehicles.
7	Smart home	Although gateways play an important role in smart homes, their centralized structure exposes them to various security risks, including integrity, confidentiality, and authorizations.	Biometrics, voice recognition, and face recognition are examples of sensitive user data kept on the Blockchain for better security. Once data is recorded on the Blockchain, it cannot be changed, and only the proper people have access to it.
8	Pharmacy industry	Blockchain can revolutionize the pharmaceutical business by introducing three key elements: privacy, transparency, and traceability.	Blockchain in the pharmaceutical supply chain can enable the precise location of medications to be identified. Batch reminders may be sent out or carried out in a timely and effective manner while ensuring patient safety.
9	Agriculture industry	Deal with climate change, soil erosion, and the loss of biodiversity.	It supports the creation and implementation of data-driven technologies for intelligent farming and index-based agriculture insurance by providing a safe way to store and manage data. Blockchain and IoT-enabled agriculture provides low cost and high revenue generations
10	Water management	Distribution of water resource to the end users is complicated because water management has traditionally relied on outdated technology and convoluted methods that may no longer be applicable today.	provide constant nature, speed, and clarity of data, allowing partners to be a part of the ongoing process rather than being beneficiaries of retrospective analysis and investigations.

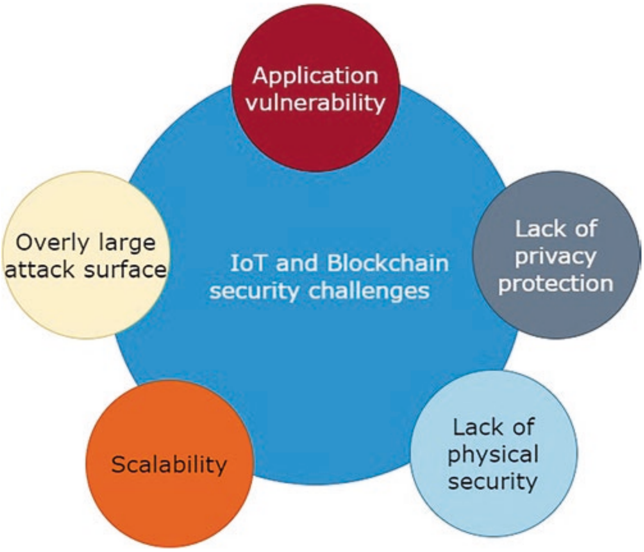

Fig. 8 Security challenges of Blockchain and IoT

might be handily stayed away from by not uncovering it. During advancement, administrations like Telnet, SSH, or an investigate interface might be helpful; however, they're seldom utilized underway.

2.11.2 Application Vulnerability

IoT devices use application to provide a service to end-users. Application contains security hole to attack the IoT devices. In uncommon conditions, the assailant might have the option to run their own code on the IoT devices, permitting them to reap delicate information or target outsiders. Security weaknesses, as other programming interface, are difficult to altogether dispose while planning programming. There are, in any case, ways of forestalling notable weaknesses or limit the probability of them happening. There are suggested rehearses for keeping away from application weaknesses, like after procedures are accepted consistently.

2.11.3 Lack of Privacy Protection

IoT devices are transferring information from one device to another. The information transmitted lacks integrity and privacy. IoT devices that are associated with a remote organization save the organization's secret phrase. Cameras can record video and sound of the space in which they are introduced. On the off chance that aggressors approached this data, it would be a significant break of protection of gadgets, and associated administrations should deal with touchy information fittingly, safely, and just with the end-authorization. client's This is valid for both the capacity and dispersal of touchy information.

2.11.4 Lack of Physical Security

IoT edge devices lack physical security. Hackers can open it and assault its equipment. Any defensive modification, for instance, can be bypassed by getting to the substance of memory parts directly. Moreover, the IoT edge devices might incorporate troubleshooting contacts that are available subsequent to opening the gadget, giving an assailant additional alternative. Actual attacks influence only one gadget and require actual contact. Since these attacks cannot be brought out in mass through the Internet, we don't believe them to be one of the most genuine security issues, yet they are as yet included.

2.11.5 Scalability

One of the most significant challenges still facing IoT devices is scalability. IoT devices deal with the huge volumes of data received by a big network of sensors, as well as the potential for slower transaction processing rates or latency. Defining a distinct data model ahead of time helps save time and avoids problems when it comes to putting the solution into production.

2.12 Blockchain and IoT Companies

Helium

Helium is a San Francisco, California-based IoT blockchain startup known as the principal decentralized machine network on the planet. The organization influences blockchain for interfacing low-power IoT machines like microprocessors and switches to the web. The blockchain-based remote web foundation of Helium is likewise one of the trademark features of the startup. With a blockchain-based framework utilizing radio innovation, Helium is one of the fruitful blockchain IoT organizations with capacities for fortifying web associations. Furthermore, the blockchain-based framework of Helium likewise helps in accomplishing critical decreases in power needed for working shrewd gadgets.

NetObjex

One more top expansion among blockchain IoT projects in the current occasions would be NetObjex. The Irvine, California-based startup has been effective in making a norm, decentralized foundation for IoT gadgets to empower correspondence between them. Moreover, the blockchain-based IoToken presented by the organization conveys a safe advanced stage to empower connection and correspondence between savvy gadgets in a single environment.

NetObjex states that the IoToken is likewise reasonable for consistent correspondence with different gadgets in a wide scope of businesses. For instance, benefactors could utilize the IoToken through their crypto-wallets to make installments for their suppers. What's more, IoToken could likewise be valuable for drone conveyance by denoting the place of conveyance and guaranteeing installment check.

Xage Security

Xage Security is one more miracle among blockchain IoT new businesses to rise out of California. It is one of the exceptional blockchain-based security stages for IoT. Curiously, the focal point of Xage Security on various businesses like transportation, horticulture, utilities, and energy demonstrates a great deal about its usefulness.

The blockchain of Xage guarantees that IoT gadgets are sealed and can get to get lines of correspondence between shrewd gadgets. Besides, Xage additionally includes a lengthy set-up of decentralized IoT applications with different capacities. You can discover applications for security strategy the executives just as ones for giving moment warnings in regards to vindictive hacking exercises.

Grid+

The Austin, Texas-based Grid+ is likewise one of the top names in blockchain IoT projects with generously valuable functionalities. Grid+ uses Ethereum blockchain for empowering customers to get energy-productive IoT gadgets. The organization's representative buys and sells power for the sake of a Grid+ client.

Moreover, the Grid+ application offers refreshed data in regards to energy utilization with the organization's savvy meter associated remotely to energy-proficient shrewd gadgets. The Ethereum blockchain of Grid+ helps specialists in paying for power at a hole of like clockwork. In particular, the principles of cutting edge blockchain cryptography enable the security of installments and general network safety on the application.

Atonomi

Atonomi is one more illustration of blockchain IoT new companies zeroed in on engaging IoT gadget security. The essential objective of Atonomi centers around expanding IoT gadget security. It perceives that the absence of gadget security hampers the improvement of IoT innovation impressively.

Atonomi offers the extraordinary functionalities of personality approval, extensible engineering, interoperability, gadget notoriety, and an exchange-based climate. Curiously, Atonomi has been customized as an installed blockchain-based answer for engaging IoT.

Riddle&Code

Riddle&Code is likewise one more fascinating expansion among IoT blockchain new companies. It offers cryptographic labeling answers for blockchain in savvy store network and coordinations of the executives. Riddle&Code brings exceptional worth from the blend of disseminated record organizations and IoT gadgets.

Thus, it could convey a coordinated, protected equipment and programming answer for empowering entrusted and secure communications with machines in IoT biological systems. In actuality, Riddle&Code fundamentally offers a “confided in computerized character” to machines or any actual gadget. The striking part of Riddle&Code alludes to the right harmony between the requirement for actual documentation and the benefits of blockchain.

HYPR

HYPR is one more conspicuous passage among blockchain IoT organizations with a great deal of possibilities. The organization uses decentralized organizations or the force of blockchain for the security of associated ATMs, homes, locks, and vehicles. Digital assaults can be incredibly wrecking because of compromises in unified information bases, which house multitudinous passwords in a single spot. HYPR follows an extraordinary instrument of putting away logins on its blockchain. Thus, it can guarantee security and decentralization of significant and touchy data.

A portion of the significant biometric security conventions of HYPR incorporate the special voice, facial, palm, and eye acknowledgment instruments for IoT gadgets. Moreover, HYPR is likewise one of the best blockchain IoT projects lately with its genuine use cases. For instance, the stage has tried biometric checks on cell phones for empowering admittance to ATM individual banking. Moreover, HYPR has likewise fostered a DLT-advanced key for mortgage holders to empower a solitary passageway for nearly everything, including IoT-empower entryways just as brilliant diversion communities.

Chronicled

Chronicled is effectively one of the top augmentations among IoT blockchain new businesses with promising potential. The stage has successfully consolidated blockchain and IoT items for conveying a start to finish production network arrangement. The essential focal point of Chronicled is by and by on the drug and the food supply businesses.

The organization uses IoT-empowered steel trailers and sensors for getting to ongoing reports with respect to the transportation cycle. The execution of blockchain in IoT gadgets in the clinical or food inventory network helps increment straightforwardness all the while. Above all, it further develops familiarity with the chain of authority for resolving any issues emerging throughout delivery.

IOTA

IOTA is fundamentally a convention to guarantee quicker exchange settlement with the worth of information respectability. It is one of the fascinating blockchain IoT projects, which includes a Tangle record for lessening the requirement for costly mining. Particle offers a productive foundation for IoT gadgets that need to manage handling of enormous measures of microdata.

The provisions of Tangle record, the blockchain record supporting IOTA, enable the abilities of the IOTA convention. A portion of the unmistakable provisions of Tangle incorporate quantum-safe information, machine-to-machine correspondence, and charge less micropayments. Besides, IOTA has additionally fostered a sensor information commercial center with plans for entering the area of information driven examination and experiences.

ArcTouch
The last expansion among the best blockchain IoT new companies in the current occasions would point towards ArcTouch. The organization represents considerable authority in planning blockchain-based programming with help for better combination with IoT arrangements. ArchTouch has fostered a scope of associated, keen arrangements, including brilliant TVs, voice colleagues, and even wearables.

The organization likewise centers around the advancement of tweaked decentralized applications (DApps) that associate with the IoT gadgets. Moreover, decentralized applications created by ArcTouch include an upgraded level of IoT security. Simultaneously, it could empower quicker handling of arrangements through savvy contracts.

3 Conclusion

Blockchain is a significant era in the recent modern computer technology and is to be implemented in different areas. Blockchain technology has found diverse applications in various sectors, including but not limited to advertising, insurance, societal use cases, energy financing, automotive, and the agriculture industry. The prior survey on blockchain addresses either technical issues or specific applications in the Internet of Things (IoT). Our paper presented the security and privacy for the Blockchain and IoT-based applications. A future direction is to take many Blockchain applications to be implemented in IoT in order to improve the security and privacy in different areas.

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